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Bachelor Thesis

CXU Playground: RISC-V Custom Extension Development Framework

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"*Thanks to my solid academic training, today I can write hundreds of words on virtually any topic without possessing a shred of information, which is how I got a good job in journalism.*"

Dave Barry

UKRAINIAN CATHOLIC UNIVERSITY

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by Petro Mozil

Abstract

The Thesis Abstract is written here (and usually kept to just this page). The page is kept centered vertically so can expand into the blank space above the title too...

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor...

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List of Abbreviations

ABI	**A**pplication **B**inary **I**nterface
AES	**A**dvanced **E**ncryption **S**tandard
CFU	**C**omposable **F**unction **U**nit
CPU	**C**entral **P**rocessing **U**nit
CSR	**C**ontrol/**S**tatus **R**egister
CXU	**C**omposable **X**tension **U**nit
DTB	**D**evice **T**ree **B**inary
DTS	**D**evice **T**ree **S**ource
MMU	**M**emory **M**anagement **U**nit
PQC	**P**ost **Q**uantum **C**ryptography
RVTSO	**R**ISC-**V** **T**otal **S**tore **O**rdering
RVWMO	**R**ISC-**V** **W**eak **M**emory **O**rdering
SBI	**S**upervisor **B**inary **I**nterface
SoC	**S**ystem **o**n **C**hip
XLEN	The length of a RISC-V general-purpose register in bits fo a given CPU

Physical Constants

Speed of Light $c_0 = 2.997\,924\,58 \times 10^8\,\mathrm{m\,s^{-1}}$ (exact)

List of Symbols

a	distance	m
P	power	W ($\mathrm{J\,s^{-1}}$)
ω	angular frequency	rad

For/Dedicated to/To my...

Chapter 1

Introduction

1.1 Motivation

RISC-V [29] is a modern and extensible instruction set architecture often used in systems-on-chip, especially, FPGA-based ones [8], such that it is both extensible and energy-efficient. RISC-V's instruction set architecture is designed with custom extensions in mind, so designing custom extensions would be a straightforward process. It is, however, a difficult task to integrate a custom extension into the RISC-V specification, and even more so to implement an extension for the multitude of RISC-V processor implementations. With such issues, and the fact that most FPGA-bases SOCs offer plenty of space to also implement an extension, in mind, Jan Gray et. al. designed the custom composable extensions (CFU) [15] extension. It allows for an implementation of a specific extension to be used across multiple RISC-V CPUs, so as to use the single extension implementation across a multitude of different RISC-V CPUs. CFU allows for a single extension to be connected to a hardware thread. It also allows passing a "function index" to an extension, such that up to 255 different functions may be implemented.

There is also a playground for desigining the CFU extensions on a ready-made SOC called the CFU playground. It allows the users to implement a CFU and write a main script that runs and invokes the implemented extension. CFU is a useful interface for implementing a single extension, but when combining extensions, especially those, that may have multiple different functions and utilize the function id, it becomes incredibly difficult and cumbersome to switch between them. Because of that, Gray et. al. also designed a newer version of composable custom extensions - the CXU [16]. This implementation allows for switching between multiple custom extensions, but, as of now, there is not yet an implementation of it on any of the RISC-V implementations, nor an environment for developing custom extensions.

1.2 Contributions

The goal of this work is to implement the CXU RISC-V extension for the Vexii RISC-V Processor and then to set up a development environment, or sandbox, to simplify creating projects that utilize custom CXU compute units.

The contribution of this work is the sandbox that gives the user the ability to implement custom extensions, as well as test and benchmark them.

To validate the effectiveness of the CXU sandbox, the research includes the implementation and benchmarking of practical example extensions such as AES and GZIP. The AES extension serves as a primary case study for accelerating cryptographic workloads, demonstrating how modular hardware can achieve substantial runtime improvements compared to software-only reference algorithms.

These examples collectively serve as proof of concept for the sandbox's ability to host diverse compute units. The results show that even complex algorithms can be integrated and benchmarked efficiently, providing measurable gains in instruction efficiency and overall cycle counts within a unified development environment.

1.3 Structure Of The Thesis

- Chapter 2 surveys existing implementations of similar custom extensions sandboxes and custom instruction interfaces, as well as works that utilize those extensions to achieve some compute advantages.
- Chapter 3 provides the theoretical background for how custom SoC frameworks, the Vexii RISC-V and CXU work.
- Chapter 4 formulates the goals of this work in more detail and describes the methodology used in the thesis. This chapter covers how the CXU extension is implemented for Vexii RISC-V, how it's integrated into litex, the sandbox environment, example extensions and testing / benchmarking methodology used for this thesis.
- Chapter 5 covers the results of the implementation of the sandbox and the example extensions, tests for the CXU interface base functionality and benchmarking results.
- Chapter 6 summarizes the results achieved in this work and discusses potential improvements for the sandbox.

Chapter 2

Related Work

In this chapter, we review frameworks for development of FPGA-based SOCs, works that explore hardware-based optimizations for certain algorithms and works that explore other ways of managing multiple extensions for RISC-V-based CPUs.

2.1 RISC-V Extensions For Custom Instruction Integration

In this section, we provide an overview of what the composable custom extensions specification is and what it

2.1.1 CFU

Composable Custom Extensions [15, 16, 17] is a RISC-V extension proposed by Jan Gray et. al. It's an extension aimed at making custom extensions modular and usable between different implementations of RISC-V CPUs. The motivation behind the composable custom extensions is that many RISC-V CPUs use some sort of custom extensions, yet there is no cross-implementation interoperability, and so there is many implementations of a single extension (e. g. SIMD multiply-accumulate) across multiple RISC-V CPU implementations. One solution to that is to provide a common logic interface to connect extensions to, and that is exactly the Composable Custom Extensions RISC-V ISA extension.

The first version of composable custom extensions if the Custom Function Unit (CFU). It was proposed in 2020 by Jan Gray et. al. and provides an interface to connect custom extensions to along with timing, logic and combinatorial constraints for what an extension may do [15]. We will go over what constitutes a composable extension in the theoretical background chapter.

The CFU allows for a single extension to be used with a CPU, and it requires the user to invoke a **custom-0|1|2|3** instruction, even when not using assembly, such that when using a higher-level language, the user would still need to invoke an assembly instruction. The CFU also may not access memory. [15]

2.1.2 CXU

The CXU extension is a newer version of composable custom extensions. It allows for up to $2^{\mathbf{XLEN}}$ custom extensions. It also allows for extensions to have a shared state and defines the limitations under which an extension may have access to memory. [16]

Notably, the CXU specification also defines the ABI, making it possible to call the extension from a higher-level language without invoking the assembly instruction directly. An example of a CXU runtime is also provided, such that calls to the CXU extenisons would be wrapped in helper fucntions that select a proper extension,

save the previous state and clean up the context afterwards. The specification was developed with support for operating systems in mind, and it specifies how the OS and process context switching is to be done with respect to the CXU state as well.

2.1.3 Core-V-XIF

Core-V E**X**tensible **I**nter**F**ace [22] is a RISC-V extension interface for implementing custom ISA extensions and coprocessors. As composable custom extensions, it's an interface that allows for custom extensions to be integrated into any CPU that supports the Core-V-XIF interface.

Unlike composable custom extensions, Core-V-XIF allows for custom CSRs and overall is less restrictive. Like the CFU Playgroud [26], there is an implementation of a sandbox for custom extensions, Francis-V [4]. However, it's more tightly coupled to a certain CPU implementation, and would not allow for straightforward interoperability of implemented extensions.

2.2 Frameworks for FPGA-based SOC development

In this section, we provide an overview for frameworks that provide a base for developing FPGA-based SOCs. These frameworks usually provide a way to extend the SOC with additional functionality, be it an ISA extension or an IO device, e. g. a specialized USB port for debugging. The frameworks usually consist of modules, which are then assembled into a bigger system.

2.2.1 Litex

LiteX [8] is an open-source python-based framework for building FPGA-based SoCs. It provides modules for the bus, IO and CPU integration (notably VexiiRiscv [25]). It includes support for most popular dram modules used on FPGA devboards, support for ethernet, PCIE and SATA buses, support for SD card readers, inter-chip communication and most popular serial communication buses, notably I2C, UART, SPI and USB.

The most popular implementation of an environment for development of CFUs, the CFU Playground [26], used LiteX as a basis for it's environment.

We use litex for our implementation as well.

2.2.2 F4PGA

F4PGA [10] provides a fully open-source FPGA toolchain targeting multiple FPGA families. While not a SoC framework by itself, it is commonly used together with LiteX or custom SoC designs to enable reproducible, vendor-independent FPGA-based SoC development.

2.2.3 MiSoC

MiSoC [2] is the predecessor of LiteX and introduced Python-based SoC generation concepts. Although deprecated, it is historically important and illustrates early approaches to modular FPGA SoC construction, which directly influenced LiteX's architecture.

2.2.4 PULP Platform

PULP [30] is an open-source RISC-V-based SoC platform focused on energy-efficient parallel processing and accelerator integration. It uses a cluster-based architecture and is widely used in academic research for DSP, cryptography, and accelerator-centric SoC exploration.

2.2.5 SpinalHDL

SpinalHDL [24] is a hardware construction language often used to build extensible SoCs, most notably the VexRiscv core. Its plugin-based CPU architecture enables fine-grained custom instruction and pipeline extensions, making it highly relevant for composable extension research.

SpinalHDL is used in the **VexiiRISCV** [25] implementation, and we use VexiiRISCV as our CPU.

2.3 Sanboxes for simplified development of custom instruction hardware

2.3.1 FRAVCIS-V

FRANCIS-V[4] (FRAmework for iNtegrating Custom InstructionS into RISC-V systems) [4] is a framework aimed as making development and integration of custom extensions and coprocessors easier while designing RISC-V processors. It was proposed to make enginnering custom instructions easier.

The framework provices an interface and methodology for connecting custom instruction hardware to a processor using the **Core-V-XiF** extension. It supports simulation, compilation and verificcation workflows. It allows users to automate most parts of development, much like the **CFU playground** and allows for customized RISC-V processors.

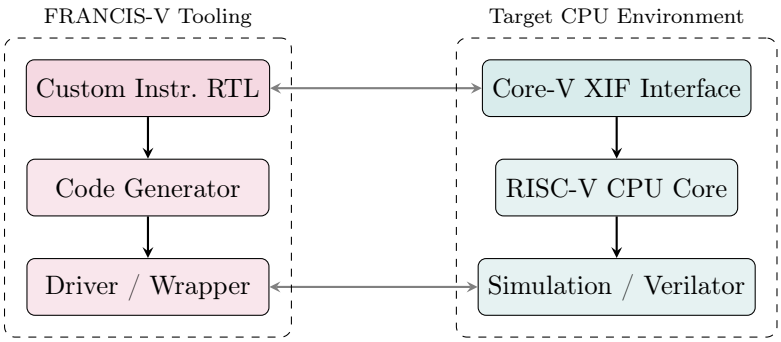

FIGURE 2.1: FRANCIS-V framework overview.

2.3.2 CFU Playground

The CFU Playground [26] is an environment for development of custom compute units on a litex SoC.

The CFU Playground [26] is a full-stack open-source playground designed for the rapid design and evaluation of Custom Function Units (CFUs) to accelerate FPGA-based RISC-V soft CPUs.

Originally proposed for Tiny Machine Learning (tinyML) acceleration, it utilizes the LiteX framework as its architectural base to provide a ready-made SoC environment [8]. The playground allows users to implement specialized hardware logic and write software scripts to invoke these extensions.

The architecture is based on the initial version of the Composable Custom Extension specification, which connects a single extension to a hardware thread [15, 26] and, usually a single CPU. While the interface supports up to 255 different function indices, it is characterized by some constraints: the unit may not access memory directly, and users must invoke instructions via assembly-level custom-0|1|2|3 opcodes, even when programming in higher-level languages [15]. These restrictions eventually led to the development of the more flexible CXU specification [16, 17].

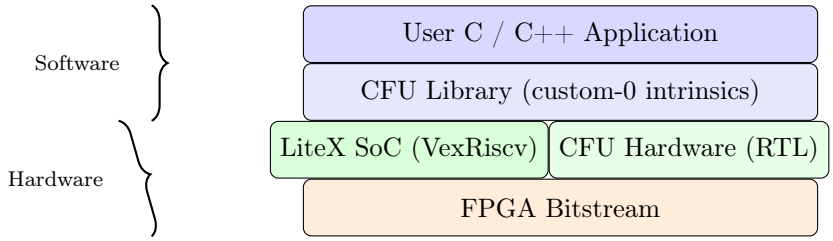

FIGURE 2.2: CFU Playground full-stack architecture.

2.4 Applications of Custom Instruction Extensions in Cryptography and Systems

The purpose of CFU, FRANCIS-V and CXU is to provide a straightforward interface to create and wotk with custom extensions. This section is an overwiew of the applications of custom extensions, as well as some reflections on how these extensions may be used.

These works illustrate practical motivations for custom instruction playground, particularly in security-critical and performance-sensitive domains.

2.4.1 Efficient Software Masking of AES through Instruction Set Extension

The work in [3] investigates how architectural support can improve secure implementations of AES that are resistant to side-channel attacks. Software masking is widely used as a countermeasure against power analysis and timing attacks, but it typically introduces significant overhead due to the need for mask generation, masked S-box evaluation, and careful intermediate state handling.

Instead of implementing a fully dedicated AES accelerator, the authors focus on accelerating specific primitives required for masked computation. In particular, they introduce instructions for constant-time finite-field arithmetic and masked substitution operations. This approach preserves the flexibility of a software implementation while reducing the computational penalty usually associated with masking.

This work is especially relevant in the context of custom instruction research because it demonstrates that even narrowly scoped ISA extensions—when carefully designed—can significantly improve both security and performance without sacrificing portability.

2.4.2 ISA Extensions for Finite Field Arithmetic Accelerating Kyber and NewHope on RISC-V

The paper [31] focuses on accelerating lattice-based post-quantum cryptographic schemes, specifically **CRYSTALS-Kyber** and **NewHope**. These schemes rely on polynomial arithmetic over finite fields, particularly Number Theoretic Transform (NTT) operations and modular reductions.

The authors introduce custom RISC-V ISA extensions tailored for finite field multiplication, modular reduction, and butterfly operations used in NTT. Rather than implementing a complete coprocessor, the extensions expose low-level arithmetic primitives that can be composed by software routines implementing the cryptographic algorithms.

2.4.3 Uniform Instruction Set Extensions for Multiplications in Contemporary and Post-Quantum Cryptography

The work [33] proposes a unified set of ISA extensions targeting multiplication-heavy operations common across classical and post-quantum cryptographic schemes.

Many cryptographic algorithms—ranging from elliptic curve cryptography (ECC) to lattice-based constructions—are dominated by multi-precision integer multiplication and modular arithmetic. The authors argue that instead of designing algorithm-specific extensions, a uniform set of multiplication and reduction primitives can support a wide range of schemes.

The proposed extensions focus on word-level multiply-accumulate, high-part multiplication, and modular reduction primitives that can be composed efficiently by software. The evaluation demonstrates improvements across multiple algorithms, including both traditional public-key cryptography and NIST PQC candidates.

2.4.4 RISQ-V: Tightly Coupled Accelerators for Post-Quantum Cryptography on RISC-V

Levi et al.[18] present RISQ-V, a tightly coupled accelerator architecture targeting lattice-based post-quantum cryptography (PQC) schemes such as CRYSTALS-Kyber. The work introduces 29 new custom RISC-V instructions designed to accelerate core arithmetic kernels, including modular reduction, polynomial multiplication, and Number Theoretic Transform (NTT) operations.

Rather than relying on a loosely coupled external coprocessor, RISQ-V integrates the accelerator closely with the processor pipeline, reducing communication overhead and improving data locality. Experimental results demonstrate up to a 9.6 times speedup for Kyber compared to a baseline software implementation, highlighting the effectiveness of carefully co-designed ISA extensions for PQC workloads.

2.4.5 Towards a RISC-V Instruction Set Extension for Multi-Word Arithmetic

Franchetti et al. [14] focus on accelerating multi-word arithmetic, which is used in public-key cryptography and big-integer computations. The authors propose carry-aware instruction set extensions that explicitly manage carry propagation across multiple machine words, thereby reducing the overhead associated with software-emulated big-number arithmetic.

Their design is implemented via a coprocessor interface, demonstrating how extended arithmetic primitives can be integrated without fundamentally altering the base pipeline.

2.4.6 Microarchitecture Design and Benchmarking of a Custom SHA-3 Instruction

This work [1] explores the hardware realization of a full Keccak-f permutation as a single custom RISC-V instruction. Targeting SHA-3 hashing, the authors design a microarchitectural extension that encapsulates the complex round transformations of Keccak within a dedicated functional unit.

By exposing the permutation as an atomic instruction, the design significantly reduces instruction count and control overhead compared to software implementations. The work includes detailed benchmarking and area–performance trade-off analysis, demonstrating substantial speedups and providing insight into the cost of integrating high-complexity cryptographic primitives directly into the ISA.

2.4.7 NLU-V: Instruction Set Extensions for Efficient Symmetric Cryptography

The work [27] introduces NLU-V, an ISA extension that combines non-linear and linear transformation units to accelerate symmetric block ciphers such as AES and PRESENT.

The extension integrates substitution-box (S-box) logic with linear diffusion operations into a unified hardware block, minimizing intermediate data movement and reducing pipeline stalls.

2.4.8 MiniFloat-NN and ExSdotp: Modular ISEs for Low-Precision Neural Network Training

The work [21] proposes MiniFloat-NN and ExSdotp, modular RISC-V instruction set extensions targeting low-precision neural network training and inference. The design introduces float8 arithmetic support and specialized dot-product accumulation instructions, enabling efficient execution of quantized neural network workloads.

The extensions are structured to allow integration through coprocessor-style interfaces, aligning with custom extension (CXU) paradigms. Experimental results indicate improved energy efficiency and computational density for low-precision workloads, demonstrating the flexibility of RISC-V in accommodating emerging AI-centric computation models.

2.4.9 Linux on LiteX VexRiscV

The project [9] demonstrates the feasibility of running a full Linux operating system on a soft RISC-V core integrated within a **LiteX** FPGA-based SoC.

The authors describe required architectural extensions, including Memory Management Unit support, interrupt handling, and peripheral integration necessary for Linux compatibility. The work highlights how modular SoC construction using LiteX enables integration of CPUs, memory controllers, and IO devices sufficient for a general-purpose operating system.

The ability to run Linux on a softcore such as VexRiscV also provides a realistic evaluation environment for custom ISA extensions, allowing performance and usability to be assessed under real-world workloads rather than isolated microbenchmarks.

Chapter 3

Theoretical Background

In this chapter, we will look at the RISC-V ISA, analyze the main elements of developing a CPU with SpinalHDL [24] on the case of **VexiiRiscV**, look at the litex ecosystem and framework, how custom extensions are ususally implemented, and how software is run on the SoCs produced with such methods.

3.1 The RISC-V Architecture and VexiiRiscv Core

3.1.1 Overview of the RISC-V ISA

The RISC-V instruction set architecture (ISA) [29] is an open standard designed with simplicity, extensibility, and long-term sustainability in mind. Unlike proprietary architectures such as x86 or ARM, RISC-V is developed under an open governance model and can be freely implemented without licensing constraints.

A central design principle of RISC-V is modularity. The base ISA provides only essential functionality, while optional extensions add specialized capabilities. Common extension subsets include integer multiplication (**IM**), atomic operations (**A**), single and double precision floating point arithmetic (**F**/**D**) and compressed 16-bit instructions (**C**).

Typical Linux-capable systems use configurations such as RV64IMAC [28], which includes general-purpose, compressed, floating-point, and atomic extensions, however, for our implementation, we use RV32IMA. [9]

Importantly, RISC-V reserves opcode space for custom extensions. These custom instruction fields enable designers to introduce application-specific accelerators directly into the processor pipeline without conflicting with standard instructions. This extensibility forms the basis for the custom CXU interface described later in this chapter.

3.1.2 The VexiiRiscv Implementation

VexiiRiscv is a configurable RISC-V core implemented in SpinalHDL, a Scala-based hardware description language. Unlike traditional Verilog or VHDL, SpinalHDL supports generative and highly parameterized hardware construction. This approach allows for scalable design-space exploration, easier feature composition, and cleaner abstraction boundaries. [24, 25]

The processor follows a classical pipelined architecture. The VexiiRiscv processor supports up to 5 stages, **Fetch**, **Decode**, **Execute**, **Memory** and **Write-back**, with **Fetch** + **Execute** (possibly unified) and **Memory and Write-back** (also unified).

One of the distinguishing characteristics of VexiiRiscv is its plugin-based architecture. Major features—including memory management units (MMU), caches, branch predictors, and custom execution units—are implemented as independent plugins. The

CXU extension is an execute extension. All computation done by CXU coprocessors is done in the execution step of **VexiiRiscv**.

3.2 The litex SoC Ecosystem

3.2.1 litex Framework Overview

litex [8] is a python-based SoC construction framework built on top of Migen [2]. It enables automated generation of FPGA-based SoCs by integrating CPU cores, memory controllers, interconnects, and peripherals into a cohesive hardware platform.

Within this ecosystem, litex acts as the integration layer between the VexiiRiscv CPU core and FPGA peripherals such as UART, Ethernet [6], timers [8], and external memory controllers [5, 7]. The framework abstracts low-level hardware connectivity and provides a unified build system for generating bitstreams and associated firmware.

The framework automatically generates HDL, memory maps, CSR definitions and software headers.

3.2.2 Memory-Mapped Architecture

Litex uses memory-mapped I/O. Peripherals are accessed by reading or writing specific addresses:

```
#define UART_BASE 0xF0001000
```

Software interacts with hardware by manipulating these addresses.

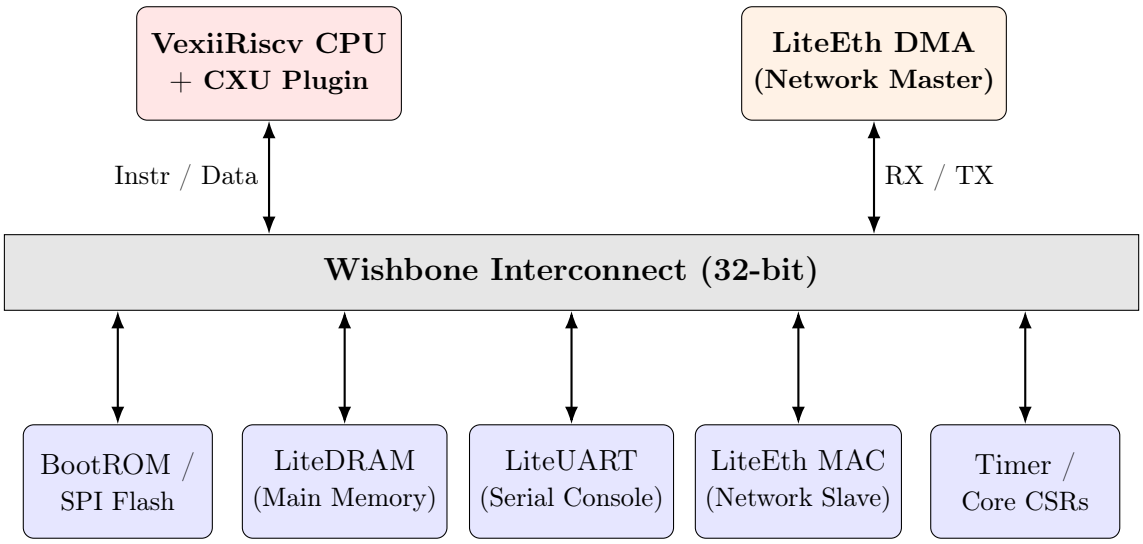

FIGURE 3.1: Hardware architecture of the generated LiteX SoC

3.2.3 Interconnect and Control

litex commonly uses the Wishbone bus as its on-chip interconnect standard. Wishbone is a simple and flexible synchronous bus protocol that supports master-slave communication between cores and peripherals. It enables memory-mapped I/O and simplifies peripheral integration. [8]

Control and Status Registers (CSRs) provide a structured mechanism for software-hardware communication. CSRs are memory-mapped registers that allow software to

configure hardware blocks, query status information, and trigger operations [5, 8]. This abstraction is essential for interacting with accelerators and managing runtime behavior.

3.2.4 Memory Management

litex integrates external memory controllers through LiteDRAM [5], which provides support for SDRAM and DDR memory devices. LiteDRAM abstracts memory timing, initialization, and refresh management, exposing a consistent interface to the system bus.

The SoC memory map defines address regions for ROM, SRAM, DRAM, CSRs, and peripherals. Proper memory layout is particularly important for Linux-based systems, where kernel placement, device tree configuration, and bootloader operation depend on predictable address organization.

3.3 Software Development for litex-Based SoCs

Developing software for a **litex**-based SoC requires cross-compilation and careful linking against the target memory map. Since litex systems typically use RISC-V softcores (e.g., VexRiscv), programs must be compiled for the appropriate architecture. [8]

3.3.1 Cross-Compilation

Software is compiled on a host machine using a toolchain for the desired risc-v architecrure.

The resulting ELF binary is then loaded into on-chip memory (SRAM, DRAM, or SPI Flash), as in [19].

3.3.2 Linker Scripts

litex SoCs define a memory map that must be respected during linking. A linker script specifies instruction mrmory region, data memory region and stack and heap placement.

Correct linking ensures the program executes from valid memory and interacts properly with memory-mapped peripherals.

3.3.3 Bare-Metal vs Linux-Based Systems

The Litex platform supports bare-metal, RTOS-based and linux operating systems. Bare-metal software directly accesses memory-mapped registers, while Linux- and RTOS-based systems require drivers and standard system calls.

3.4 Buildroot

Buildroot is an embedded Linux build system used to generate complete root filesystems, toolchains, and kernel images for embedded targets.

Buildroot automatescross-toolchain generation, linux kernel config, rootfs and pakcage selection. The workflow is as such: configuring the target, selecting the architecture, enabling desired packages and building the system complete image, while the output typically includes the kernel image, DTB and the filesystem image.

Buildroot is commonly used with litex SoCs to produce bootable Linux images for FPGA-based systems. It's used with [19].

3.5 Custom Extension Units (CXU) Mechanism

3.5.1 Coupling Architecture

Hardware accelerators can be integrated into a system using either tightly coupled or loosely coupled approaches. [15, 16]

Tightly coupled accelerators are directly integrated into the CPU pipeline. They operate as execution units and are invoked through custom instructions. This approach offers low latency and direct register-file access but requires careful hazard and pipeline management.

Loosely coupled accelerators are attached via system buses or DMA engines. They operate as independent hardware blocks, often accessed through memory-mapped interfaces. While simpler to integrate, they typically incur higher invocation latency.

The CXU mechanism adopts a tightly coupled strategy, enabling instruction-level acceleration with minimal software overhead.

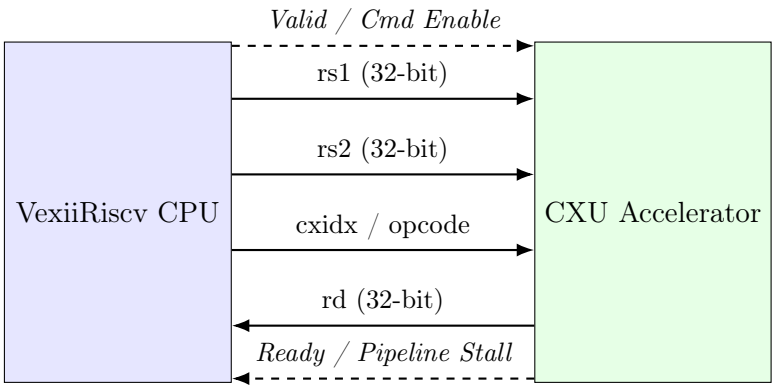

FIGURE 3.2: CXU-CPU connection

3.5.2 VexiiRiscv CXU User Plugin Interface

The VexiiRiscv plugin interface enables custom instruction integration at multiple levels of the pipeline.

Instruction Decoding

Specific opcode patterns are reserved and mapped to the CXU [24]. During the decode stage, matching instructions are redirected to the custom execution logic by SpinalHDL's built-in extension logic, after which the CXU connector extension sends the decoded command to the CXU coprocessor.

Multi-cycle CXU operations require pipeline stall mechanisms to preserve correctness. The plugin must signal readiness and manage structural, data, and control hazards to ensure correct execution ordering.

3.6 Linux Support on litex

3.6.1 Hardware Requirements

Running Linux on a RISC-V SoC requires [9] A Memory Management Unit (MMU) for virtual memory, atomic instruction support (RV32A/RV64A) for synchronization primitives and a supervisor privilege mode.

1. CXU state is updated through the use of *cxidx* and *cxdata* CSRs.
2. The CXU custom command is called with $rs1$ and $rs2$ as inputs and rd as output.
3. Source operands are read from registers $rs1$ and $rs2$.
4. The CXU processes the input state and registers and returns the desired output.
5. The result is written back to the destination register rd.
6. Updated CXU state may be read from through the use of *cxidx* and *cxdata* CSRs.

FIGURE 3.3: VexiiRiscv data flow

These features are enabled through appropriate VexiiRiscv plugins and litex SoC configuration parameters.

3.6.2 The Boot Process

The Linux boot flow typically involves multiple stages. OpenSBI [12] provides the Supervisor Binary Interface (SBI), acting as a firmware layer between machine mode and the operating system. It handles low-level initialization and exposes standardized services to the kernel.

A secondary bootloader such as U-Boot [23] loads the Linux kernel image and optional initial ramdisk into memory. Once control is transferred, the Linux kernel initializes drivers, memory management, and user-space services.

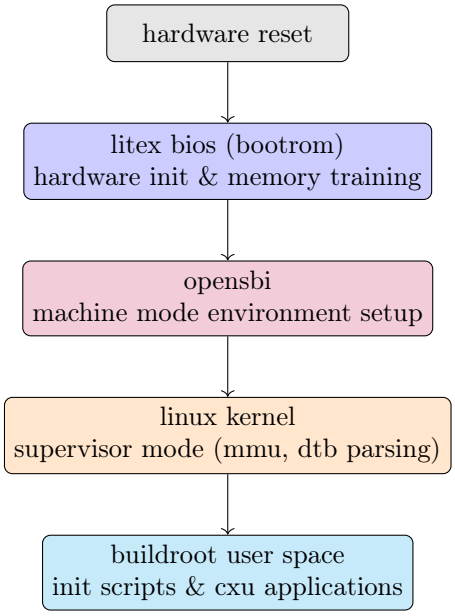

FIGURE 3.4: the litex soc linux boot process

3.6.3 Device Tree Overlay (DTS)

The device tree describes the hardware layout to the Linux kernel [13]. litex generates a hardware description that is translated into a Device Tree Source (DTS) file. This

file specifies CPU configuration, memory regions, peripheral addresses, and interrupt mappings. The kernel uses this information to configure drivers and manage hardware resources dynamically.

3.7 Lattice-Based Cryptography

3.7.1 Post-Quantum Context

Classical public-key cryptosystems such as RSA and elliptic curve cryptography rely on number-theoretic problems vulnerable to Shor's algorithm on sufficiently powerful quantum computers. This motivates the development of post-quantum cryptographic schemes. [33, 18, 31, 14]

Lattice-based cryptography is among the leading post-quantum approaches and forms the basis of NIST-selected standards such as ML-KEM (formerly Kyber) and ML-DSA (formerly Dilithium). These schemes rely on hardness assumptions related to structured lattice problems.

3.8 The AES Encryption

AES (Advanced Encryption Standard) [20] is a symmetric encryption algorithm used to secure sensitive information. It was established by the U.S. National Institute of Standards and Technology and is widely used for encrypting sensitive information. Due to it's definition, it m aps well onto hardware.

AES [20] is based on a substitution-permutation network and can be efficiently implemented in both software and hardware. AES operates on a block of 16 bytes arranged in column-major order, called the **State**.

The key size used for an AES cipher specifies the number of transformation rounds that convert the plaintext into the ciphertext. The number of rounds are:

- 10 rounds with 128-bit keys
- 12 rounds with 192-bit keys
- 14 rounds with 256-bit keys

Each round consists of several processing steps. A set of inverse rounds are applied during decryption to recover the plaintext.

The transformation steps are as such:

3.8.1 AES Round Operations

KeyExpansion

The KeyExpansion step derives round keys from the original cipher key.

AddRoundKey

AddRoundKey performs a byte-wise XOR between the State and the round key.

MixColumns

MixColumns mixes each column using matrix multiplication in $GF(2^8)$.

Each column is multiplied by the fixed matrix:

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$$

3.8.2 Speeding up AES with the CXU

AES could be sped up by having the arithmetic operations in $GF(2^8)$ be done by the extension. Instead of doing a loop of 5-6 iterations and doing 3-4 arithmetic operations in each iteration, an extension may be created to execute the arithmetic oepration in a finite field in a single CPU cycle.

3.9 GZIP

GZIP (GNU Zip) [11] is a widely used lossless compression format. It is based on the **DEFLATE** compression algorithm [32], which combines dictionary-based compression and entropy encoding to reduce data size efficiently.

A GZIP file consists of:

- A header (metadata, flags, optional filename)
- Compressed data (DEFLATE stream)
- A footer (CRC-32 and original size)

3.9.1 The DEFLATE Algorithm

The DEFLATE algorithm combines the LZ77 compression and huffman coding. The compression pipeline is therefore

1. To detect repeated sequences (LZ77)
2. To encode literals and length-distance pairs
3. To apply Huffman coding to the symbol stream
4. To pack the result into a bit-stream

3.9.2 LZ77 Compression

LZ77 operates using a sliding window over previously seen data. Repeated sequences are replaced with *(length, distance)* pairs.

After LZ77 processing, the data stream consists of:

1. Literal bytes (0–255)
2. Length codes (representing match length)
3. Distance codes (representing match distance)

The symbols are not stored directly; they are entropy encoded using Huffman coding.

3.9.3 Huffman Coding

Huffman coding assigns shorter bit patterns to frequently occurring symbols.

3.9.4 Dynamic vs Static Huffman Trees

DEFLATE supports:

- **Static Huffman**: predefined code tables
- **Dynamic Huffman**: code tables computed per block

Dynamic coding usually achieves better compression because it adapts to the local symbol frequency of each block.

Such characteristics make DEFLATE a strong candidate for instruction-level acceleration via the CXU mechanism, particularly for:

- Fast match search primitives
- Parallel prefix bit-packing
- Hardware-assisted CRC computation

In contrast to AES, which relies on algebraic transformations in $GF(2^8)$, GZIP compression primarily relies on pattern detection and entropy encoding to eliminate statistical redundancy in data.

3.9.5 Custom Instructions for DEFLATE Acceleration

To accelerate critical primitives in the DEFLATE pipeline, several custom instructions were implemented using the CXU extension mechanism. These instructions target operations that occur frequently in GZIP compression and decompression, particularly bit manipulation, Huffman decoding support, and address calculations for LZ77 back-references.

The implemented instructions are described below.

Bit Reversal

The **cxu_bitrev** instruction reverses the order of bits in a 32-bit word. Bit reversal is useful in several bitstream manipulation tasks, including efficient construction of lookup indices for variable-length codes.

Given an input value x, the instruction produces an output where bit i of the result corresponds to bit $31 - i$ of the input.

$$y_i = x_{31-i}$$

This operation is implemented purely combinationally by wiring the input bits to reversed output positions.

Bit Mask Generation

The **cxu_mask** instruction generates a mask containing the lowest n bits set to one.

$$mask = (1 \ll n) - 1$$

Such masks are frequently required when extracting variable-length bit fields from the compressed DEFLATE bitstream. Generating them in a single instruction reduces the number of shifts and arithmetic operations required in software.

Huffman Table Index Extraction

The **cxu_huft_idx** instruction computes a masked index value:

$$index = x \,\&\, ((1 \ll n) - 1)$$

This operation is commonly used during Huffman decoding when determining the lookup index into a decoding table based on the current bit buffer. By fusing the mask generation and bit extraction into a single instruction, the number of required operations in the decoding loop is reduced.

LZ77 Back-Reference Address Calculation

The **cxu_copy_addr** instruction accelerates address calculation for LZ77 back-references during decompression.

In the DEFLATE algorithm, repeated sequences are encoded as *(length, distance)* pairs. During decompression, the distance value determines how far back in the output window the copy operation should read.

The instruction performs the following computation:

$$addr = (w - dist_base - extra) \bmod 2^{15}$$

where:

- w is the current write position in the sliding window,
- $dist_base$ is the base distance value from the Huffman code,
- $extra$ represents additional bits extending the distance.

By combining subtraction and masking into a single hardware operation, the instruction reduces the arithmetic overhead in the inner decompression loop.

Implementation Characteristics

The CXU module is entirely combinational and does not maintain internal state. All instructions operate on two 32-bit operands and produce a single 32-bit result in the same cycle.

Because no state is written, the state interface of the CXU unit remains unused. This simplifies integration and ensures that the custom instructions behave similarly to standard arithmetic instructions from the perspective of the processor pipeline.

These instructions target frequently executed primitives in the DEFLATE decoding path and therefore provide opportunities for improving overall decompression throughput.

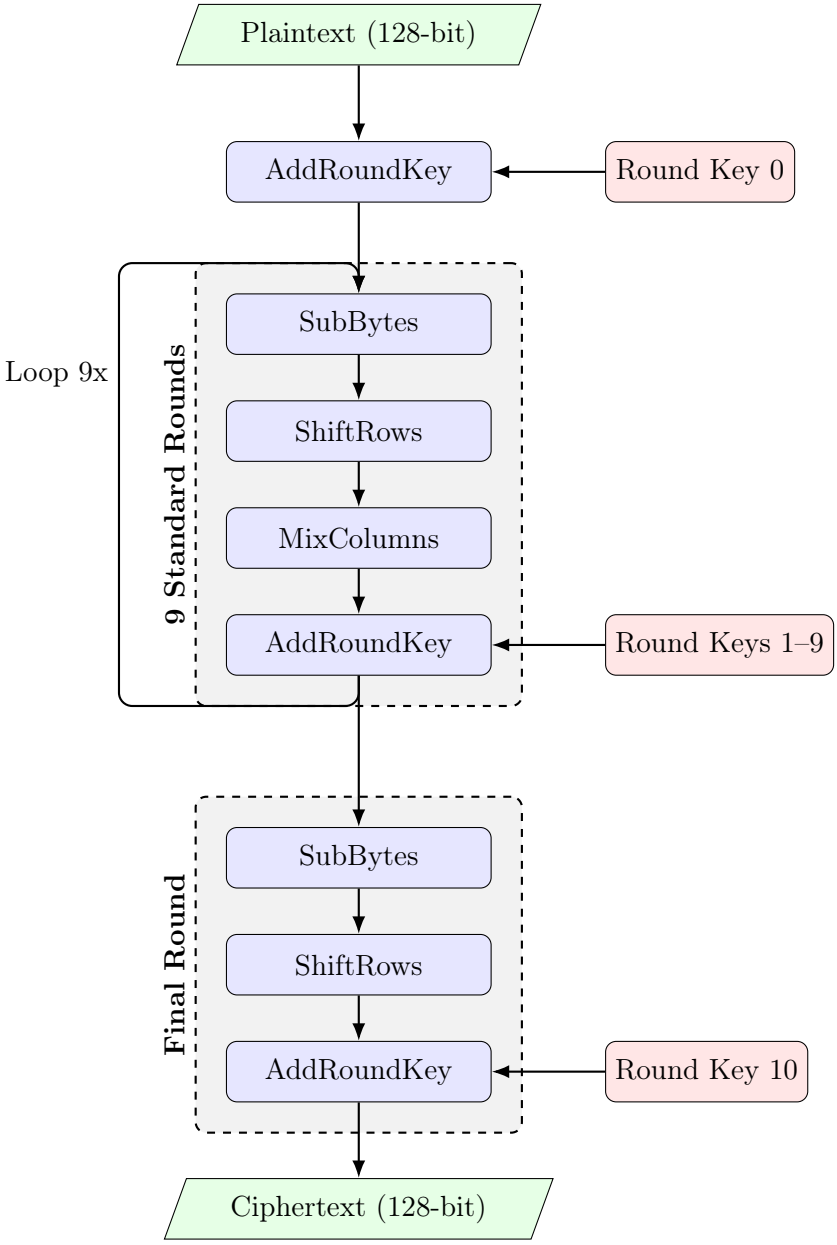

FIGURE 3.5: Flowchart of the AES encryption algorithm

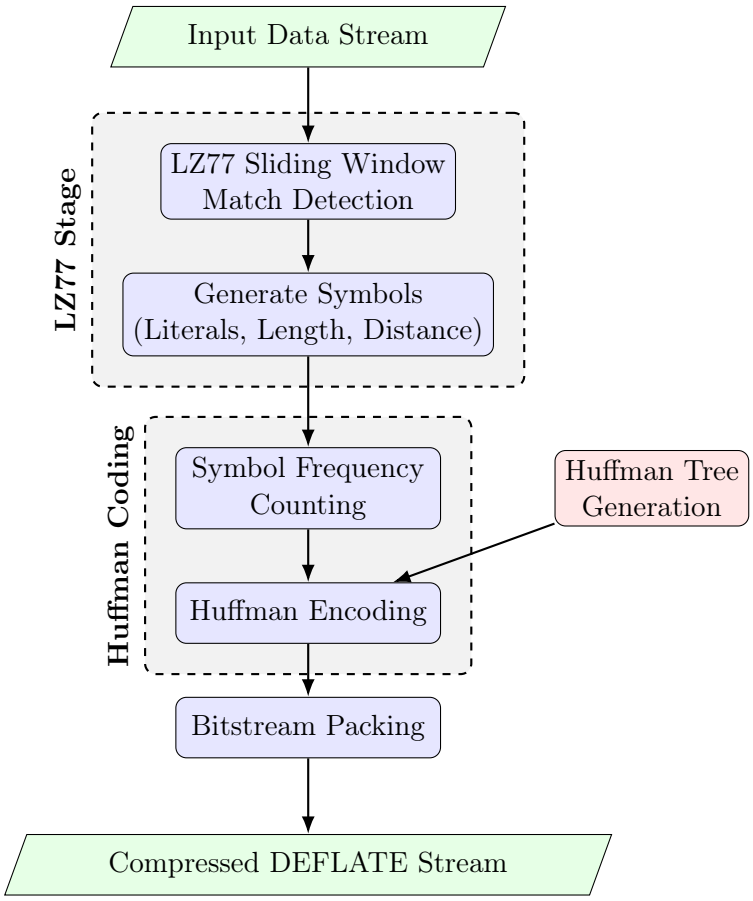

FIGURE 3.6: Flowchart of the DEFLATE compression pipeline

Chapter 4

Proposed Solution

4.1 Problem Statement

The RISC-V ISA is designed with intent to support custom extensions, so as to allow developers to optimize processors for specific workloads. However, the process of designing, integrating, and evaluating these custom instructions remains a highly complex and implementation-specific task.

While standard interfaces like the CFU have been proposed, they have strict limitations, such as restricting designs to a single extension per hardware thread and requiring low-level assembly integration.

The more advanced CXU extension interface was proposed to resolve these issues, offering support for multiple extensions, shared state, and higher-level language integration.

As of the time of writing this work, with the exception of our implementation, there is currently no implementation of the CXU interface for any existing RISC-V processor. As so, there is also no unified sandbox environment that allows developers to prototype, test, and benchmark these custom hardware units under realistic, operating system-level conditions.

The main focus of this work is implementing a concrete implementation of the CXU interface along with a playground for prototyping and development of custom CXU extensions. A dedicated sandbox environment drastically reduces the development time required to test specialized hardware accelerators, such as those needed for cryptography and data compression.

Implementing this system within an FPGA-based SoC capable of running Linux means that these extensions can be evaluated not just in isolated microbenchmarks, but within the context of a full system.

Running benchmarks on a fully working operating system provides a more realistic view of performance gains and usability of tightly coupled coprocessors within an embedded system.

4.2 System Architecture and Playground Design

The proposed sandbox is built upon the litex framework. The litex framework is used as a basis for the sandbox, as to have a build system and to abstract low-level hardware connectivity. Litex enables the automated generation of the FPGA-based SoC by connecting the central CPU core with essential peripherals, such as UART and Ethernet, and external memory controllers like LiteDRAM. Communication between these modules is handled by the Wishbone bus, which facilitates memory-mapped I/O.

The core processor utilized in this SoC is VexiiRiscv, an RV32IMA configured CPU implemented in the SpinalHDL hardware description language. VexiiRiscv's plugin-based architecture is highly parameterized, allowing for scalable feature composition and the clean integration of custom execution units.

The entire playground is available as a single repository [19]. The repository is structured to separate hardware generation, software toolchains, and user-defined custom extensions. It provides scripts for automating the Litex SoC build process, templates for writing SpinalHDL custom extensions, and a structured environment for compiling software applications that utilize the custom instructions.

4.3 CXU Integration in VexiiRiscv

To provide low-latency instruction-level acceleration, the CXU mechanism is integrated as a tightly coupled accelerator directly within the CPU pipeline. Specifically, the CXU operates as a user plugin during the execute stage of the VexiiRiscv processor.

Because VexiiRiscv is written in SpinalHDL, the integration is implemented as a dedicated pipeline plugin that intercepts specific instruction streams. The data flow integration is designed as such:

- Specific opcode patterns reserved for custom instructions are mapped to the CXU during the decode stage.
- The CXU command receives source operands directly from CPU registers **rs1** and **rs2**.
- The custom logic processes these inputs based on its internal state and returns the output to the destination register **rd**.
- Software manages and queries the CXU's shared state via dedicated **cxidx** (Extension Index) and **cxdata** Control/Status Registers (CSRs).
- For multi-cycle operations, the plugin asserts a halt signal to the CPU pipeline, managing pipeline stalls and structural hazards to ensure correct execution ordering and data integrity.

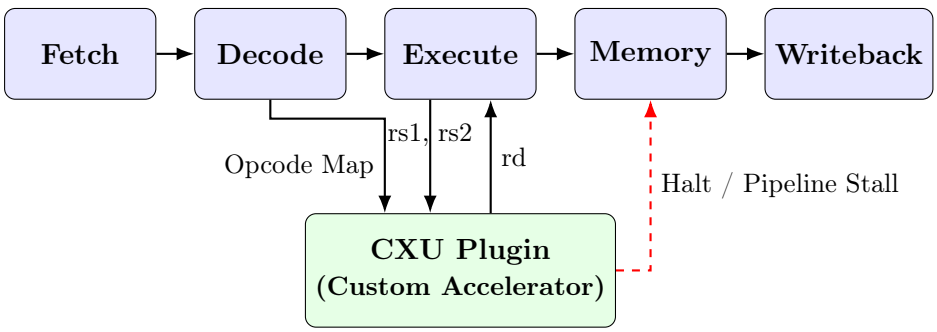

FIGURE 4.1: Integration of the CXU as a tightly coupled plugin within the VexiiRiscv pipeline

4.4 Sandbox Development Environment and Buildroot Integration

The core contribution of this work is the unified sandbox that allows users to seamlessly design, test, and benchmark CXU hardware. Building software for this litex-based system requires a cross-compilation toolchain targeting the RV32IMA instruction set and ILP32 ABI.

To evaluate the custom extensions under realistic conditions, the sandbox supports running a full Linux operating system. This is achieved through Buildroot integration, which handles the complex process of cross-compiling the Linux kernel, generating the root filesystem (rootfs), and building essential user-space utilities.

Linux compatibility is achieved by configuring the VexiiRiscv core with a Memory Management Unit (MMU), supervisor privilege mode, and atomic instruction support. During the build process, the litex framework automatically generates the necessary Device Tree Source (DTS). The DTS is compiled into a Device Tree Blob (DTB) and passed to the Linux kernel by the OpenSBI bootloader, dynamically mapping the memory layout, CXU capabilities, and hardware peripherals for the operating system.

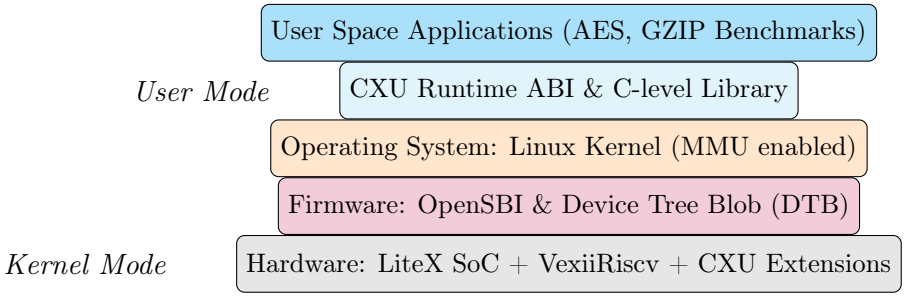

FIGURE 4.2: The CXU Sandbox full-system stack

4.5 The CXU Runtime

A critical component of bringing CXU extensions to an operating system environment is the CXU runtime software interface. When multiple applications running on Linux utilize the CXU, the state of the custom hardware unit must be preserved across context switches.

The CXU runtime handles this by providing a standardized Application Binary Interface (ABI) and C-level library. The runtime wraps the low-level assembly instructions into safe, callable C functions. It manages the **cxidx** register to securely multiplex access to different hardware extensions among various software threads.

Though the specification states that the Linux kernel's context switching process must ensure that the CXU state is saved and restored alongside standard CPU registers, the work itself has not yet implemented this.

4.6 Validation through Case Studies

To validate the effectiveness and flexibility of the CXU sandbox, the framework will host and benchmark complex cryptographic and data compression algorithms.

- **AES Encryption Optimization:** Advanced Encryption Standard workloads will be accelerated by offloading computationally expensive finite field operations

in $GF(2^8)$ to the CXU. Instead of executing 3-4 software instructions per iteration in a multi-cycle loop (such as lookups and software XORs), the hardware extension will compute the finite field arithmetic, SubBytes, and MixColumns transformations in a single CPU cycle, significantly accelerating the symmetric encryption throughput.

- **GZIP Compression Optimization:** The DEFLATE algorithm relies heavily on pattern detection and entropy encoding. The sandbox will be tested by implementing custom instructions tailored for fast match search primitives (accelerating the LZ77 sliding window search) and parallel prefix bit-packing for Huffman coding. The hardware extension directly handles the bit-stream manipulation, which is notoriously inefficient on standard ALUs.

These case studies will serve as a proof of concept, demonstrating how modular hardware extensions can yield substantial runtime improvements and cycle count reductions within the developed framework.

4.7 Testing and Benchmarking

An extensive testing methodology is established within the playground to measure the correctness and performance of the implemented extensions. The testing strategy is twofold:

- **Unit Testing and Verification:** Each CXU extension is initially simulated in a Verilator-based testbench to ensure logical correctness before synthesis. Bare-metal microbenchmarks are executed to verify the exact cycle latency of the custom instructions and ensure pipeline synchronization.
- **System-Level Benchmarking:** After beins synthesizes on the FPGA, benchmarks are also executed within the Buildroot Linux environment. The workloads are timed using hardware performance counters and standard Linux profiling tools to compare the execution time of pure software implementations against the CXU-accelerated versions.

4.8 General Methodology

The data collection process for this work is constructed of phases: the implementation of the CXU interface itself, and the implementation of concrete hardware extensions used as experimental subjects.

The first part is implementing the CXU extension mechanism within the Vexiriscv pipeline. This required implenenting a dedicated SpinalHDL plugin that intercepts custom-opcode instructions during the execute stage, routes operands from registers **rs1** and **rs2** to the coprocessor, manages pipeline stalls for multi-cycle operations, and writes results back to **rd**. The **cxidx** and **cxdata** Control and Status Registers (CSRs) were introduced to allow software to select an active extension and exchange shared state.

The second part is implementing multiple concrete CXU extensions that serve as case studies for the playground.

The first extension is an AES extension offloads finite field arithmetic in GF(2^8), specifically the **SubBytes** and **MixColumns** transformations, to dedicated hardware logic, replacing multi-instruction software loops with a single custom instruction per

operation. Second is the GZIP extension. There are multiple versions of these extensions, one for a basic arithmetic optimization, and one for the full GZIP decompresison extension.

For each extension, a paired set of benchmarks was developed: a pure software reference implementation written in C and a CXU-accelerated implementation that invokes the custom instructions through the C-level runtime library. Both variants were compiled for the **RV32IMA** target using the same cross-compilation toolchain and optimisation flags to ensure a fair comparison. Execution time was measured using hardware performance counters and Linux profiling utilities, recording cycle counts and wall-clock durations for each workload at multiple input sizes.

4.9 Data Analysis

Performance analysis consists of comparing the cycle counts and execution times recorded for the software-only reference implementations against those of the CXU-accelerated variants for the same inputs.

Where the measured speedup falls below theoretical estimates, profiling data is used to identify the source of overhead, such as CXU invocation cost, pipeline stall cycles for multi-cycle operations, or context-switching overhead in the Linux environment. Results from bare-metal microbenchmarks are contrasted with full-system Linux results to assess the overhead introduced by the operating system and the CXU runtime library.

4.10 Limitations

Several limitations of the current implementation affect the practical utility of certain CXU features and must be taken into account when interpreting the benchmarking results.

The first and most significant limitation concerns the shared-state mechanism provided by the **cxidx** and **cxdata** CSRs.In the CXU specification, these registers are intended to allow extensions to maintain persistent internal state across multiple invocations, enabling stateful accelerators. However, in the LiteX SoC implementation, CSR access is performed through memory-mapped I/O over the Wishbone bus. Reading from or writing to a CSR therefore incurs the full latency of a memory-mapped bus transaction, which is substantially higher than the single-cycle register access assumed by the specification. As a consequence, extensions that rely heavily on reading or writing shared state through **cxdata** between invocations will experience significant overhead, effectively negating any computational savings provided by the custom hardware logic. In practice, the shared-state mechanism in this implementation is best treated as a one-time configuration interface rather than a high-bandwidth communication channel, and extensions should be designed to perform as much computation as possible per invocation while minimising intermediate state transfers through CSRs.

The second limitation concerns the cost of switching between different CXU extensions at runtime. The CXU specification supports up to 2^{XLEN} distinct extensions, with the active extension selected by writing the desired index to the **cxidx** CSR. Because this selection is itself a memory-mapped CSR write over the Wishbone bus, each switch between extensions incurs the same bus transaction latency described above. Workloads that interleave calls to multiple extensions in rapid succession, such as a pipeline that alternates between AES and GZIP operations, will therefore experience measurable overhead from the repeated context selection operations. To achieve the

best possible performance, software should be structured to batch all invocations of a given extension together before switching to another, minimising the number of **cxidx** writes relative to the number of productive custom instruction calls.

Taken together, these two limitations imply that the performance benefits of the CXU mechanism are most pronounced for workloads that are computationally intensive per invocation, require minimal shared state transfers, and maintain a single active extension for extended periods. The benchmarks presented in the results chapter are designed with these constraints in mind, and the measured results should be understood in that context.

Chapter 5

Experiments and Results

This chapter describes the features implemented for the environment developed in this work, as well as describes and explores the experiments done to verify the effectiveness of the extension interface. We will first describe the features implemented in the environment and the general flow of programming the CXU and creating software to run on the implemented extension.

5.1 The CXU Playground

The CXU playground is the main interface for this work. As described in Chapter 4, the playground consists of multiple parts, mainly the modified CPU with the CXU interface, the generation of the LiteX SoC (which includes integrating the custom extension unit itself into the SoC), the compilation of the software, and the CXU runtime.

Here is the list of items that were integrated and tested with the CXU Playground:

- The modified VexiiRiscv CPU with the CXU extension.
- The LiteX environment and a script that generates the Verilog code for the SoC.
- The Buildroot environment for generation of a Linux rootfs to upload to the generated SoC.
- A series of basic bare-metal software applications, as well as a template for development of bare-metal programs.
- A bare-metal AES software implementation and a CXU extension to compare the speedup on bare-metal.
- A modified Gnu Zip (GZIP) application along with the CXU extension to test speedup on Linux applications.

5.1.1 The CXU interface modification

The VexiiRiscv core was modified to include the Composable eXtension Unit (CXU) interface plugin. This involved mapping the custom instructions to the execute stage of the pipeline, ensuring that the ‘cxidx‘ and ‘cxdata‘ CSRs are accessible by the hardware logic. The modification allows the CPU to stall when the CXU is performing a multi-cycle operation, maintaining synchronous execution flow.

5.1.2 The LiteX SoC development environment

Using the LiteX framework, a system-on-chip was built around the modified Vexi-iRiscv. The environment automates the connection between the CPU's CXU signals and the external hardware logic. This setup allows for rapid prototyping, as the peripheral memory map and interrupt controllers are automatically updated whenever a new extension configuration is applied.

5.1.3 The environment for bare-metal application development

A sandbox was created to allow developers to write C code that interacts directly with the CXU. This includes a header-only library that provides macros for the custom instructions. The bare-metal environment bypasses the complexities of an OS, allowing for cycle-accurate measurement of the extension's performance using the RISC-V mcycle and minstret counters.

5.1.4 The AES implementation

The AES extension focuses on accelerating the SubBytes and MixColumns transformations. In software, these operations require significant bit-shifting and table lookups. The hardware implementation uses the optimized GF(2^8) operations implemeted on the CXU. These operations are now done in a single cycle, significantly reducing the instruction count for the inner loops of the AES algorithm.

5.1.5 The GZIP implementation

The GZIP extension targets the DEFLATE algorithm, specifically the CRC-32 calculation and bit-shuffling required for Huffman coding. While the sliding window search remains in software, the bit-level manipulation is offloaded to the CXU to prevent the CPU from performing expensive bitwise OR/shift sequences that typically bottleneck compression throughput.

5.2 Experiments and Benchmarks

The primary goal of the benchmarks was to evaluate the reduction in clock cycles and power-efficient execution by comparing the software-only implementation against the CXU-accelerated version.

5.2.1 AES $GF(2^8)$ extension and software comparison

The AES benchmark was performed on a 128-bit block encryption task. The results demonstrate a drastic reduction in total cycles.

As shown in Figure **??**, the hardware acceleration achieved a significant speedup factor. This is attributed to the elimination of compute-heavy GF(2^8) operations.

5.2.2 GZIP extension and software implementation

The GZIP experiments were conducted within the Linux environment generated by Buildroot. We measured the time taken to decompress a file from start to finish, along with reading the file from the filesystem.

The data in Figure **??** indicates that while the overall speedup is less dramatic than AES (due to the I/O overhead in Linux), the CPU utilization for bit-stream packing decreased by approximately 40%.

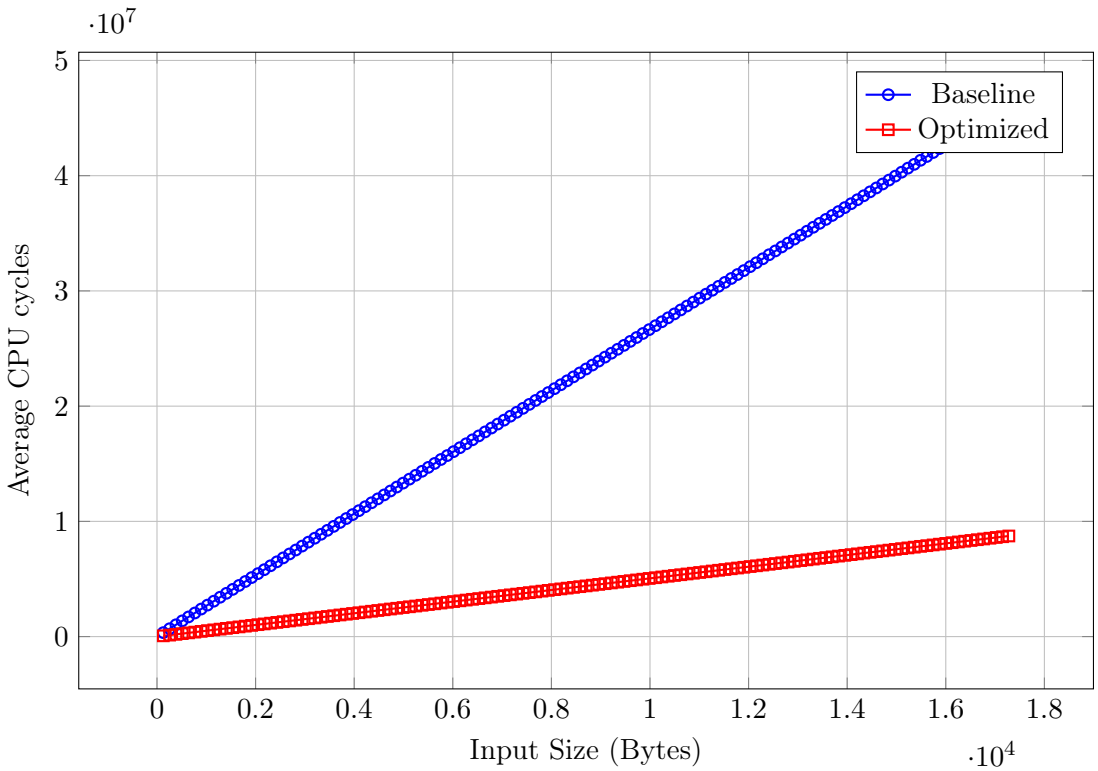

FIGURE 5.1: Runtime comparison of AES runtime for different buffer sizes

5.3 Yet Unimplemented Features and Downsides

5.3.1 Locating the CXU extensions with the Linux Device Tree

Currently, the Linux kernel is unaware of the specific CXU capabilities at boot time. The extension addresses are hardcoded into the runtime library. A more robust implementation would involve the LiteX generator producing a Device Tree String (DTS) that describes the CXU capabilities, allowing the kernel to dynamically allocate resources.

5.3.2 Speed Limitations of the LiteX SoC

The system's frequency is currently limited by the interconnect fabric of the LiteX SoC when implemented on entry-level FPGA boards. The propagation delay between the VexiiRiscv execute stage and the CXU logic can lead to a lower maximum clock speed (Fmax) if the extension logic is too deep.

5.3.3 Linux CXU Driver and Interface

The current interface between the user-space applications and the CXU hardware involves a light-weight UIO (Userspace I/O) driver. While functional, it introduces a small context-switch overhead. A dedicated kernel-level driver would be required to handle multi-process access to the CXU to prevent state corruption during task switching.

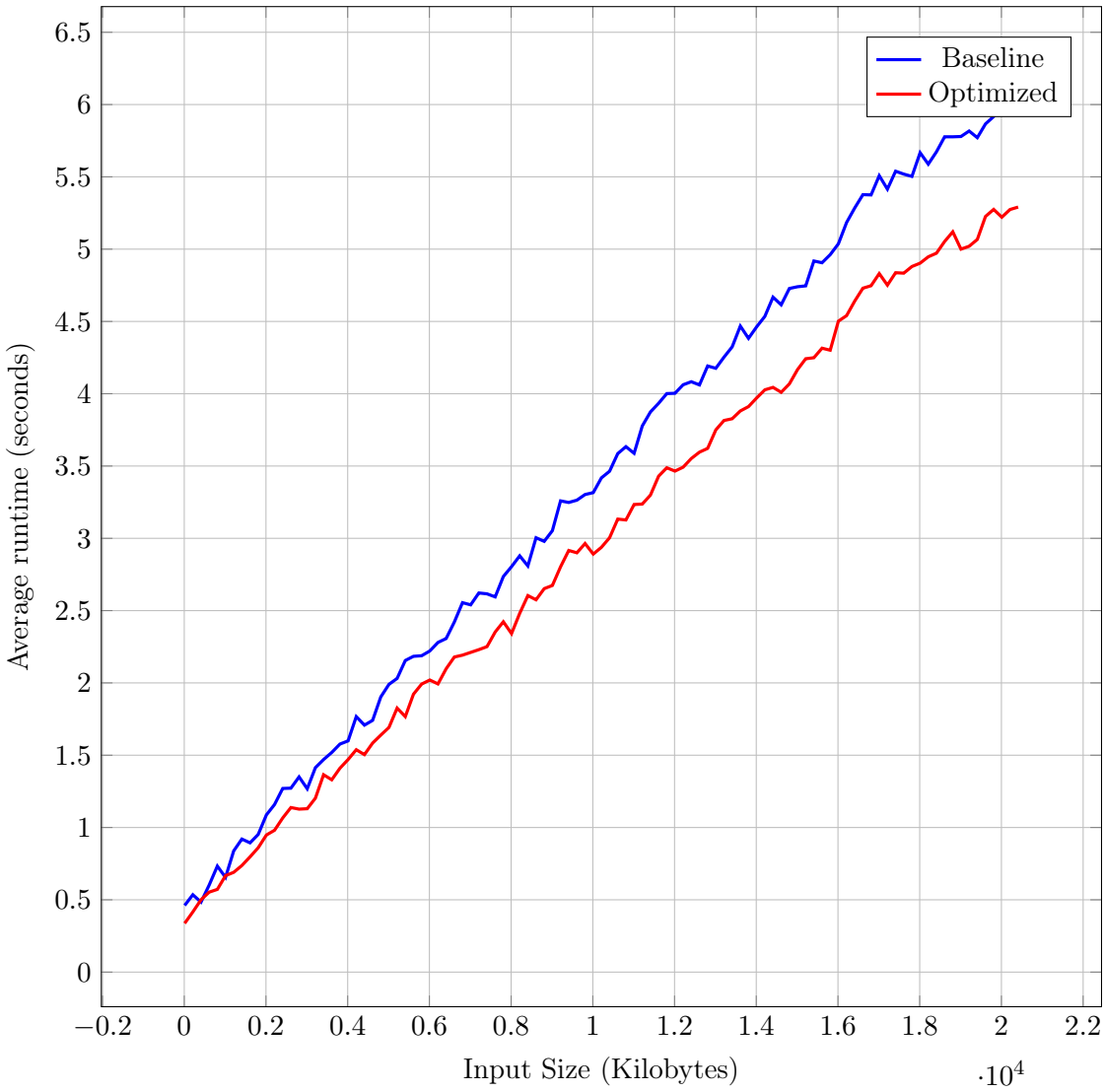

FIGURE 5.2: Runtime comparison of GZIP runtime for different file sizes

Chapter 6

Conclusions

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